MILITARY SPECIFICATION

VIEWER/PRINTER, MICROFICHE (24X AND 48X)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for - (24X and 48X) fiche viewer/printer (V/P) (see 6.4.8) capable of enlarging and projecting a fiche (see 6.4.1) microimage (see 6.4.6) onto a viewing surface, and provide a hard copy (paper) print of the image (see 6.4.4). The fiche V/P shall be capable of enlarging 24X and 48X microimages recorded in a standard 105 by 148 mm, approximately 4" x 6" fiche format.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

W-C-596	-	Connector, Plug, Electrical Connector, Receptacle, Electrical.
PPP-B-636	-	Box, Fiberboard.
PPP-T-60	-	Tape: Pressure-Sensitive Adhesive, Waterproof, for Packaging.
PPP-T-76	-	Tape, Pressure-Sensitive, Adhesive, Paper, (for Carton Sealing).

MILITARY

MIL-P-116	-	Preservation-Packaging, Method of.
MIL-C-675	-	Coating of Glass Optical Elements.

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures for Inspection by Attributes.
MIL-STD-129	-	Marking for Shipment and Storage.
MIL-STD-130	-	Identification Marking of US Military Property.
MIL-STD-147	-	Palletized and Containerized Unit Loads 40" x 48" Pallets, Skids, Runners, or Pallet-Type Base.
MIL-STD-150	-	Photographic Lenses.
MIL-STD-167	-	Mechanical Vibrations of Shipboard Equipment.
MIL-STD-454	-	Standard General Requirements for Electronic Equipment.
MIL-STD-461	-	Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-462	-	Electromagnetic Interference Characteristics, Measurement of.
MIL-STD-810	-	Environmental Test Methods for Aerospace and Ground Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein.

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AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI-PH3.46-1971	-	Projected-media Temperature Test for Photographic Still-Projection Equipment.
ANSI-PH5.10-1969	-	Method for Measuring the Screen Luminance of Microfilm Readers with Translucent Screens.
ANSI-S1.4-1971	-	General Purpose Sound Level Meters, Specification for.
ANSI-X4.12-1970	-	Office Appliances and Business Equipment, Safety Standard for (UL 114 October 1970).

(Standards may be obtained from ANSI, Inc., 1430 Broadway, New York, NY 10018.)

Microcopy Resolution Test Chart.

Test charts may be obtained from the following or equal companies specializing in microphotography:
Photographic Sciences Corporation, 23 W. Main Street, Webster, NY 14580.
Itek Corporation, Optical Systems Division, 10 Maguire Road, Lexington, MA 02713.

3. REQUIREMENTS

3.1 First article inspection. V/P's furnished under this specification shall be products that have been tested and have passed the first article inspection (see 6.3).

3.2 Material. The material for each part shall be as specified herein. However, when a definite material is not specified, a material shall be used which will enable the V/P to meet the performance requirements of this specification. Acceptance or approval of any constituent material shall not be construed as a guaranty of the acceptance of the finished product.

3.3 Design and construction. The design and construction of the fiche V/P shall be as specified herein.

3.3.1 Viewer/printer. The fiche V/P shall be designed for table or desk use and be capable of displaying and printing data recorded on the microform (see 6.4.5). V/P's shall be designed for use in office and industrial areas and shall have protective base mountings to prevent damage to furniture or other surfaces on which V/P's may be placed. The design shall not preclude permanent mounting.

3.3.1.1 Construction. The V/P shell may be constructed of painted or corrosion resistant metal, of noncombustion supporting fiberglass or high impact plastic. The shell may be used as a support member of all internal mechanisms and shall be sufficiently rigid to permit moving the viewer without damage or misalinement of optical components. All mechanisms and electrical connections shall be enclosed, but shall be readily accessible for adjustments and maintenance.

3.3.1.2 Primers/enamel. All metallic surfaces shall be corrosion resistant. Metallic surfaces when painted shall be covered with one or more coats of metallic base primer or prepared for enamel by phosphatizing metal in lieu of primer and one or more coats of synthetic enamel. Primer and finish coats shall be applied in accordance with normal industrial painting practice. The finish coat when dry shall present a smooth, even surface free from runs, sags, peels, chips, blisters, scratches, and be free of thin coating or no coating areas.

3.3.1.3 Condition. All exterior surfaces of fiche V/P's shall be dry and free from oil, grease, dirt, burrs, sharp edges, and rust. All V/P units shall be clean, well made and free from defects which affect the serviceability, and detract from its appearance.

3.3.1.4 Size. Dimensions shall be such as to permit passage through a circular hatch 25 inches in diameter and through a rectangular doorway 20 inches wide. The V/P may exceed these dimensions if it is capable of being readily disassembled by nontechnical personnel into a maximum of four parts (excluding nuts, bolts, washers, etc. used in

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holding V/P parts together) which will pass through these openings and be readily reassembled by nontechnical personnel, in which case appropriate instructions for such disassembly and reassembly shall be provided with each V/P as part of the maintenance manual.

3.3.1.5 Weight. The total weight of the V/P shall not exceed 120 pounds.

3.3.1.6 Hood. The V/P may be equipped with a hood to block overhead ambient light.

3.3.2 Viewing surface. The V/P shall be equipped with a nonglare, viewing surface capable of producing an image with good contrast and legibility. The usable viewing surface shall be at least 11 inches high and 14 inches wide.

3.3.2.1 Design. The viewing surface shall be free of scintillation and be such that the reflection of the ambient lighting (see 3.4.1), shall not interfere with the legibility of the image.

3.3.2.2 Impact resistance. The viewing surface shall be shatter-resistant and sufficiently scratch resistant to withstand normal cleaning (as specified by V/P manufacturer) without damage.

3.3.2.3 Ease of removal. The viewing surface shall be designed to be removed and replaced in less than 10 minutes without the use of special tools when removal is necessary.

3.3.2.4 Line marker. The V/P shall be equipped with a 1.0 ±0.25 mm horizontal line marker at mid viewing surface to facilitate retrieval of data on a single line.

3.3.3 Lamp.

3.3.3.1 Accessibility. The lamp shall be easily accessible for replacement within 5 minutes without requiring the use of special tools.

3.3.3.2 Lamp. The lamp shall have a rated life of at least 500 hours when operated in the V/P and in accordance with the lamp manufacturer's derating curves. The lamp shall be a regenerative cycle type with a low voltage (50 volts maximum) filament.

3.3.4 Ventilation.

3.3.4.1 Cooling blower. If a forced air cooling blower is used to remove the heat emitted from the lamp, the discharge vent must be away from the operator position. The blower must be provided with sealed lifetime lubricated bearings and be essentially maintenance free.

3.3.5 Optical system (see 6.4.7). The image on the viewing surface shall be right reading and sufficiently free of vibration, distortion, spherical aberration and chromatic aberration to the extent required to meet the requirements specified herein. Design shall permit moving V/P from one desk or table top to another without resulting in misalinement of the optical path.

3.3.5.1 Optical system accessibility and replacement. Lenses, mirrors, and other optical components shall be easily accessible so cleaning can be accomplished within 10 minutes and replacement within 30 minutes (exclusive of cooling time). Optical components shall be mounted in a way that realinement is not required after cleaning or replacement when reasonable care is taken.

3.3.5.2 Optical schematic. A schematic diagram of the optical system shall be included in the maintenance manuals required by 3.14.

3.3.6 Lenses. The lenses (24X and 48X) shall be anastigmat and the field shall be flat and shall be sufficiently corrected for spherical aberration, coma, astigmatism, and distortion to the extent required to meet the other requirements specified herein.

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3.3.6.1 Lens coating. Each air-glass surface of the projection lens shall be coated with antireflection material in accordance with MIL-C-675.

3.3.6.2 Lens marking. Each lens barrel shall be marked with the magnification of the lens and its focal length.

3.3.6.3 Magnification. The V/P shall provide for linear magnification of 24X ±8 percent and 48X ±4 percent. The entire information area within a fiche frame (see 6.4.2) (page equivalent) shall be fully displayed to allow reading all information on the viewing surface. The projected image shall be of flat field sufficient to meet the required resolution, without refocusing, except when switching from one magnification to another.

3.3.6.4 Focus. The range of focus shall be sufficient to provide a sharp legible image from either side of a 0.002 inch to 0.009-inch-thick 24X or 48X fiche. The focus control shall permit defocusing on either side of the sharpest focus. When the viewer image is in focus at its center, the four corners and the midpoints of the four edges of the image also shall be in focus.

3.3.6.5 Focus mechanism. The focus mechanism shall be essentially free of backlash. Design shall permit operator convenient focusing. The image shall remain in focus when fiche is moved from frame to frame.

3.3.6.6 Lens interchangeability. V/P's shall provide for manually operated mechanical shifting between 24X and 48X magnification without tools. Lenses shall be affixed to the V/P to require their removal by partial disassembly of the V/P with commonly available tools, if required.

3.3.7 V/P brightness control. Means shall be provided by a control located on the lower one-fourth of the V/P on the side facing the operator to allow decreasing the brightness level by 20 to 35 percent, to compensate for different film optical densities, ambient light conditions, or the operator's preference. No flicker shall be observed during operation of the V/P.

3.3.8 Fiche carrier. Each V/P shall be equipped with a vibration free carrier for retaining the 0.002 to 0.009-inch-thick fiche precisely in the optical plane while viewing projected microimages on the viewing surface. It shall provide means for positive alinement so as to minimize a canted image on the viewing surface. Once positioned, the carrier shall remain in place unless deliberately moved.

3.3.8.1 Carrier design. Fiche carrier mechanisms that require opening and closing for insertion or removal of fiche shall be selfopening, remain open, and be selfclosing for insertion or removal of the fiche. The carrier mechanism may be front or side loading, and it shall be designed for convenient operation. The carrier shall accommodate 105 mm x 148.00 mm (+0.00 -1.0 mm) fiche (see figure 1). When platens are used, a means shall be provided for loading and unloading fiche from the carrier without scratching or otherwise damaging the fiche, and arranged so all surfaces of the platen may be cleaned. If transparent platens are used, they shall be scratch resistant.

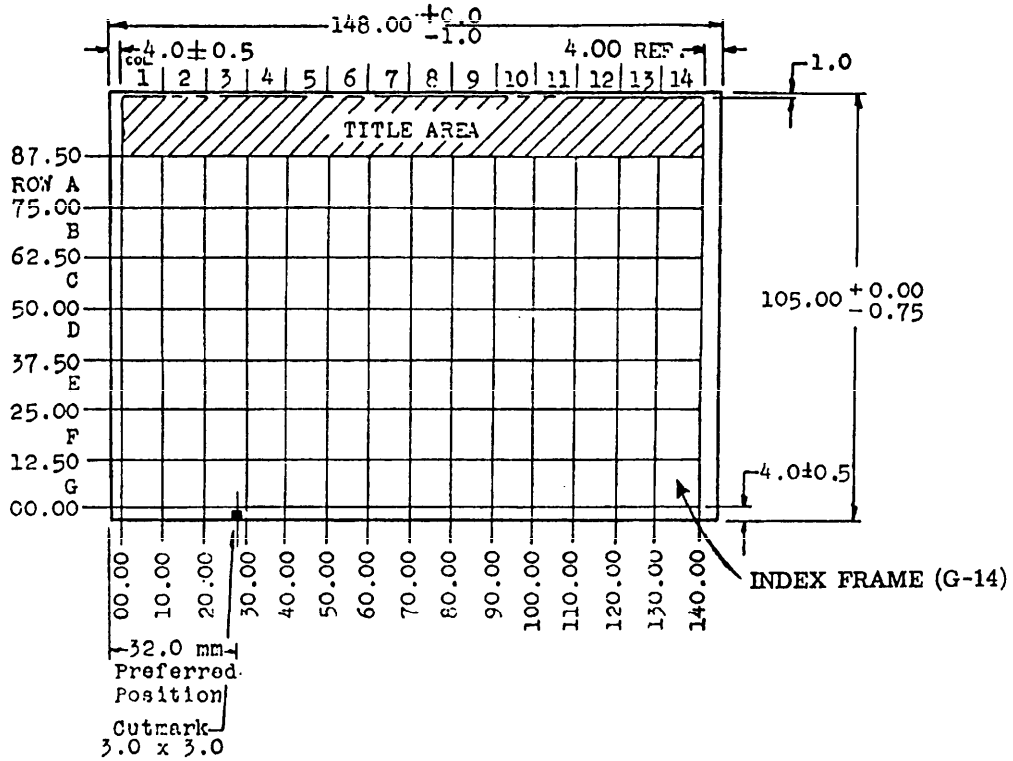
3.3.8.2 Positioning and index control. X Y positioning indexes shall be provided to assist location of specific frames within both the 98 and 270 frame format (see figure 1). The X Y indexes shall be identified on a grid (see 6.4.3) or scale so that a conveniently positioned locator control on the front of the V/P can position the corresponding frame from the fiche onto the viewing surface. Grid and grid material shall be scratch resistant.

3.3.8.3 Retaining apparatus. A positive retainer or latch shall be provided to prevent accidental removal of the fiche carrier and damage to platens if used.

3.3.9 Printer.

3.3.9.1 Printing process. A dry or semidry print process shall be used. If a semidry process is used, the printout shall be dry to the touch. The printout image shall be stable, readable, and essentially flat. Print paper shall not present an electrical shock hazard.

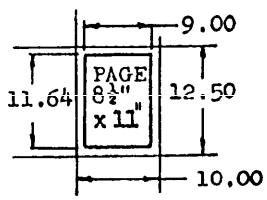
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Typical imaginary

Document

Size 8 1/2" x 11"
Text size 7.0" x 10.67"
64 lines
70 characters

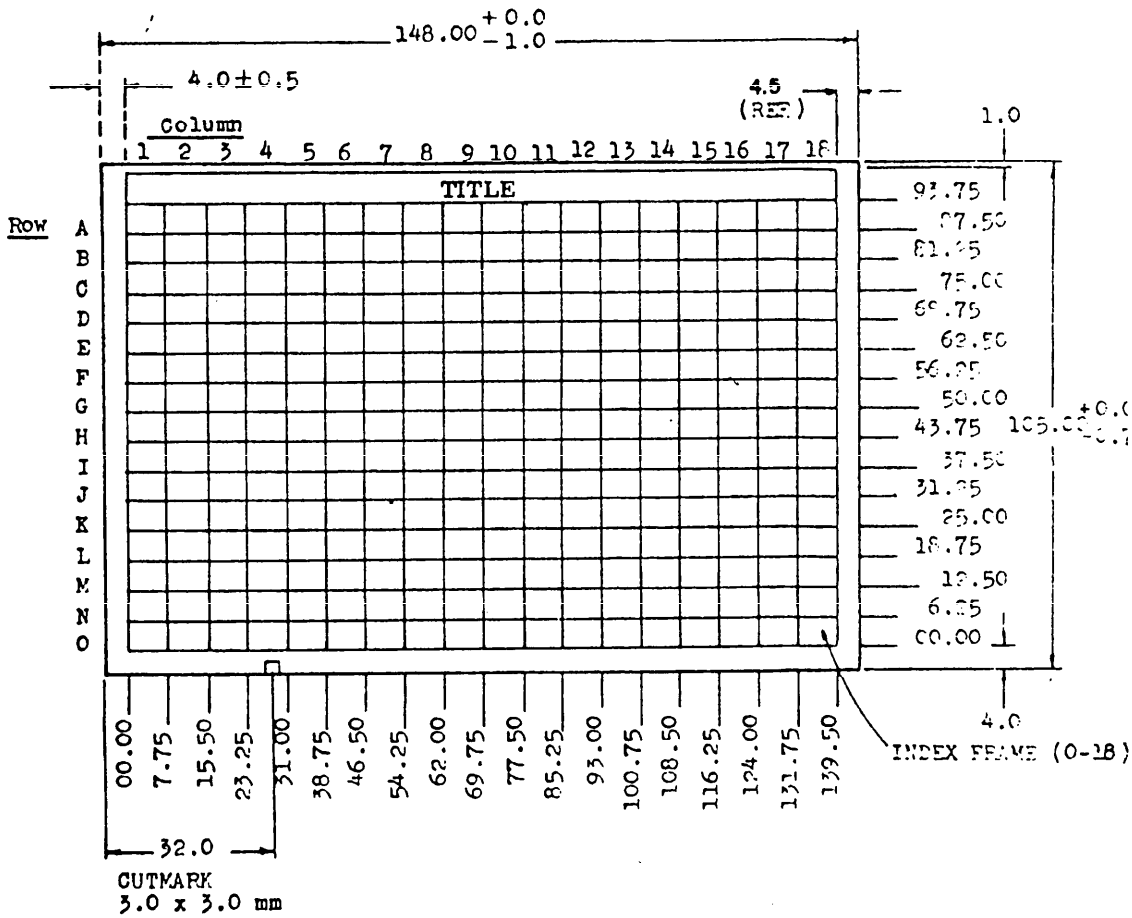


NOTES:
1. Format-14 columns x 7 rows = 98 frames.
2. Effective reduction 24X ±5%.
3. Dimensions in millimeters, except where noted.
4. Grid lines shown do not appear on microfiche.

24X

FIGURE 1. Fiche format.

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NOTES:
1. Page size 14 x 11 in. (text 13.20 x 10.67 in.)
2. Effective reduction 48X.
3. Format 18 columns x 15 rows. 270 frames.
4. Dimensions are in millimeters, except as noted.
5. Grid lines shown do not appear on microfiche.

48X

FIGURE 1. Fiche format.

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3.3.9.2 Print paper processing unit. The processing unit required for developing the exposed print paper shall be designed to prevent chemicals/toner, if required, from spilling, contaminating each other, or in any way becoming free from the processing unit and contaminating, soiling, or damaging any part of the V/P to a maximum of 30 degrees tilt from the vertical in any direction when in operation. The complete image on print produced shall be on 11-inch wide paper and a blowback of not less than 71 percent of the 48:1 microimage or 92 percent of the 24:1 microimage.

3.3.9.3 Loading. The V/P shall be designed so that paper stock can be loaded or removed in less than 2 minutes. Chemicals/toner (including their containers) shall not require more than 5 minutes to load and unload in the V/P.

3.3.9.4 Print paper feed. The equipment shall be motorized to move paper automatically from the paper storage through processing, and deliver a finished cut print within the reach of the operator's position.

3.3.9.5 Paper cutter (not applicable to precut paper). An automatic switch shall be designed into the V/P to activate a paper cutter that will cut the roll paper stock cleanly and squarely.

3.3.9.6 Time cycle. The maximum print cycle for paper movements, exposing, processing, and delivering a finished print shall not exceed 20 seconds when operating at 60 Hz, 24 seconds when operating at 50 Hz.

3.3.10 Printing material storage. The printing processing compartment shall be designed to enclose paper and chemicals to prevent deterioration of such material within the V/P.

3.3.11 Operator controls.

3.3.11.1 Print. An identified push button switch shall be provided to activate the print cycle. The location shall be convenient for both right and left-handed persons.

3.3.11.2 Exposure control (if required). An identified exposure control located on the lower one-fourth of the V/P on the side facing the operator shall be provided which shall be easily adjustable over a wide range of exposures. It shall be manually adjustable to compensate for variations of input film density of 0.10 to 0.20D minimum and 1.3 to 1.90D maximum and shall automatically control the printout exposure. The exposure control mechanism shall be inactive during film projection to the viewing surface.

3.3.12 Paper print. The printed image shall be right reading and shall be positive (black image on white background) from negative fiche inputs. When cost per printout is a bid factor, it shall be included in the contract or purchase order (see 6.2).

3.3.12.1 Legibility and contrast. The exposed and processed print shall be of maximum uniform contrast between image and background. It must be a reproduction of the image projected on the viewing surface and all data so projected must be legible and readable on the print.

3.3.13 Electrical. V/P's shall operate satisfactorily on 115 volts (±10 percent) and 230 volts (±10 percent), 50 and 60 Hz, single phase power supplies. No soldering or unsoldering shall be required to change from one voltage or frequency to the other. Unless otherwise specified (see 6.2), all viewers shall be delivered ready for operation on 115 V, 60 Hz.

3.3.13.1 Power cord. V/P's shall be equipped with at least an 8-foot long electrically insulated, 3-conductor (green ground) commercial type power cord, type SJ, or proven equivalent, UL approved capable of conducting the rated current of the V/P and have a 300 volt rating. A 2-pole, 3-wire grounding commercial type plug in accordance with W-C-596, style D series, shall be attached.

3.3.13.2 On-off switch. A front located (on the side facing the operator) on-off switch shall control the power to the V/P. In the off position, the switch shall interrupt both sides of the electrical power. The cooling blower, if required, may be connected

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so that it will automatically stop running after the viewer has cooled.

3.3.13.3 Interlocks. Interlocks shall be incorporated to eliminate the possibility of shock to operating and maintenance personnel. An interlock shall be provided which shall disconnect all power to the viewer when the shell is opened for servicing or changing lamps and fuses, except for low hazard circuits of 30.0 volts rms (line voltage) 42.4 volts peak ac (reference ANSI-X4.12-1970). Appropriate warning signs shall be installed at strategic points indicating the presence of high voltage or a hazardous condition.

3.3.13.4 Circuit breaker/fuse. Circuits shall be suitably protected by a circuit breaker or fuse not to exceed 15 amperes to prevent damage to the V/P from electrical power surges, short circuit, or overload.

3.3.13.5 Electrical schematic. Two schematic diagrams of the elctrical circuitry shall be provided as follows: One affixed to the interior of the shell in an accessible location and the other included in the technical manual. Coding of electrical circuits is required to distinguish multiple circuit patterns. The V/P's shall be adequately marked in a prominent location to indicate the voltage these units are wired for.

3.4 Operating environment. The V/P shall meet the requirements of this specification when operating in the following environmental conditions.

3.4.1 Ambient lighting. V/P's shall be designed to be used under artificial white light of 60-foot candles at desk top level.

3.4.2 Temperature. V/P's shall operate satisfactorily in temperatures from +35° to +115°F.

3.4.3 Film plane temperature. When V/P's are tested as specified in 4.8, there shall be no fiche damage, distortion, or film plane temperature in excess of 167°F.

3.4.4 Humidity. The V/P shall be designed to operate in ambient relative humidity up to 95 percent and shall meet test requirements specified in 4.9. After the tests, the V/P shall meet the requirements of 3.10.

3.5 Resolution of fine detail (see 6.4.9). When the V/P is tested as specified in 4.10, the viewer shall be capable of resolving 3.2 lines/mm and the printed shall resolve 2.5 lines/mm of a NBS 1010a/1963 target. Resolution shall be read in accordance with MIL-STD-150.

3.6 Acoustic noise. When V/P's are tested as specified in 4.11, the acoustic noise generated shall not exceed 60 dB above the threshold of hearing (0) on the A scale of a sound level meter conforming with ANSI-S1.4-1971.

3.7 Electromagnetic interference (not applicable to on-off switch). When V/P's are tested as specified in 4.12, they shall not radiate or be susceptible to electromagnetic energy.

3.8 Viewing surface brightness. When V/P's are tested as specified in 4.13, the luminance of the center of the viewing surface shall be 50-foot lamberts minimum. Luminance fall-off from center to corners A, B, C, and D, as defined in ANSI-PH5.10-1969, shall not exceed 80 percent.

3.9 Viewing surface image contrast. When the V/P is tested as specified in 4.14, the viewing surface shall have an image contrast ratio no less than 10:1.

3.10 Viewer/printer operating test. When tested as specified in 4.15, each control shall function properly, and the projected test fiche on the viewing surface and printed copy shall be easily read.

3.11 Vibration. V/P's shall either be shock mounted or otherwise designed and constructed to withstand mechanical vibrations when tested as specified in 4.16. After the test, the V/P shall meet the requirements of 3.10.

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3.12 Marking. V/P's shall be identified in accordance with the requirements of MIL-STD-130. This identification shall be located on the V/P shell.

3.12.1 Identification plate. The item identification plate shall contain the following:

(a) Item name
(b) Serial number
(c) Procurement instrument identification number
(d) Operating voltage and frequency
(e) Manufacturers' FSCM, name or trademark
(f) Federal stock number
(g) Government ownership designation

3.12.2 Identification of subassemblies and parts. Subassemblies and parts shall be marked in accordance with "Parts" and "Sub-assemblies (minor assemblies)" of MIL-STD-130.

3.13 Workmanship. V/P's shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454.

3.14 Operating and maintenance manuals. Operating and maintenance manuals shall be furnished with the V/P (see 6.5).

3.15 Repair parts. One replacement lamp and two replacement fuses (when used) shall be furnished with each V/P. Replacement lamps and fuses shall be supplied in a protective accessory pack to insure unbroken delivery at destination. The lamps and fuses shall be of standard commercial production. Replacement parts shall be identified by the actual manufacturer's part number.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspections specified herein are classified as follows:

(a) Materials inspection (see 4.3).
(b) First article inspection (see 4.5).
(c) Quality conformance inspection (see 4.6).

4.3 Materials inspection. Materials inspection shall consist of certification supported by verifying data showing that the materials used in fabricating the V/P are in accordance with the applicable referenced specifications or requirements prior to such fabrication.

4.4 Inspection conditions. All inspections shall be performed at room ambient temperature and humidity, or as specified. Test voltage and frequency shall be 115 volts, 60 Hz, unless otherwise specified.

4.5 First article inspection. Two samples shall be inspected and tested in accordance with table I and examined for defects as listed in table II. Failure to pass any tests or a major defect as listed in table II shall be cause for rejection of the first article sample.

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TABLE I. First article inspection.

Examination or test	Requirement paragraph	Method paragraph
Visual and mechanical examination		
Design and construction- - - - - - - - -	3.3 thru 3.3.9.5 incl, 3.12 and 3.13	4.7
Operating environment		
Film plane temperature 1/- - - - - - - -	3.4.3	4.8
Humidity 2/- - - - - - - - - - - - - - - -	3.4.4	4.9
Electrical mechanical		
Resolution of fine detail- - - - - - - -	3.5	4.10
Acoustic noise - - - - - - - - - - - - - -	3.6	4.11
Electromagnetic interference 2/- - - - - -	3.7	4.12
Viewing surface brightness - - - - - - -	3.8	4.13
Viewing surface image contrast - - - - -	3.9	4.14
Viewer operating test 1/ - - - - - - - -	3.10	4.15
Vibration 2/ - - - - - - - - - - - - - - -	3.11	4.16

1/ One of the 2 first article samples shall be used to perform these tests at 50 Hz.
2/ One first article sample to be submitted to these tests.

TABLE II. Classification of defects.

Examination or test	Defects	Classification	
		Major	Minor
Finish	Any rough or sharp edges or burrs.		X
	Surfaces blistered, peeled, chipped, or area of no film or thin film (metallic coatings).		X
Design and construction general	Any wearing part not accessible for adjustment, replacement, or servicing.	X	
	Any component fractured, dented, or not in proper alinement.		X
	Functional component that is inoperative.	X	
	Not adjusted to perform the function intended.		X
Threaded fasteners	Missing, broken, stripped, fractured or loose.		X
Rivets(where applicable)	Missing, broken, or loose.		X
Electrical assembly	Wiring cut, abraded, not properly joined, loose at terminal, or not inclosed where required.		X
	Adequate slack not provided for wiring to relieve strain or excessive insulation stripped from wiring.		X
Power cord	Size or length not as specified.		X
	Plug not as specified.		X
Electric circuit	No fuses or circuit breakers.		X
	Electrical requirements not as specified.	X	
Lamp	Rated life not as specified.		X
Interlocks	Misadjusted, inoperative.	X	
External index grid system	Missing - Not compatible with subindexing codes.	X	
Focus mechanism	Excessive backlash	X	

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TABLE II. Classification of defects - Continued.

Examination or test	Defect	Classification	
		Major	Minor
Focus control	Not located as specified.		X
Name plate	Missing, illegible, not as specified.		X
Lens marking	Omitted or not as specified.		X
Lens coating	Omitted. Not as specified.	X	 X
Optical system accessibility	Not as specified	X	
Operator controls	Not identified. Not located as specified.		X X
On - Off switch	Not located as specified. Not identified.		X X
Line marker	Missing, not as specified.		X
Instruction manual	Missing, incomplete, not legible.		X
Resolution of fine detail	Not as specified.	X	
Acoustic noise	Not as specified.	X	

4.6 Quality conformance inspection.

4.6.1 Inspection of product for delivery. Inspection of product for delivery shall consist of the inspections shown in table III.

4.6.1.1 Inspection lot. An inspection lot shall consist of all V/P's of the same model, size, style and design produced under essentially the same conditions, and offered for inspection within a 1-week production period.

4.6.1.1.1 Sampling plan. Statistical sampling and inspection shall be in accordance with MIL-STD-105 for general inspection level II. The acceptable quality level (AQL) shall be as specified in table III. Major and minor defects shall be as shown in table II.

4.6.1.1.2 Rejected lots. If an inspection lot is rejected, the supplier may rework it to correct the defects, or screen out the defective units, and resubmit for reinspection. Resubmitted lots shall be inspected using tightened inspection. Such lots shall be separate from new lots, and shall be clearly identified as reinspected lots.

4.6.2 Classification of defects. Examination and tests of the end item shall be in accordance with the classification of defects specified in table II.

4.6.3 In process examination. Inspection shall be performed at appropriate intermediate stations to determine compliance with all requirements which cannot be determined on the fabricated end item.

4.6.4 Preventive maintenance. The only preventive maintenance permitted during first article and production acceptance tests is the replacement of lamps or fuses that have opened because of a part failure.

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4.6.5 Disposition of sample units. Units that have been subjected to inspection of first article tests may be shipped provided the units meet the performance tests specified in 4.6. A new lamp shall be furnished with each unit (see 3.3.3.2).

TABLE III. Quality conformance inspection.

Examination or test	Requirement paragraph	Method paragraph	AQL (defects per hundred units) Major	AQL (defects per hundred units) Minor	No. per lot to be inspected
Visual and mechanical examination					
Design and construction	3.3 thru 3.3.9.5 incl, 3.12 and 3.13	4.7	2.5	6.5	See 4.6.1.1.1
Electrical mechanical					
Resolution of fine detail	3.5	4.10			
Acoustic noise	3.6	4.11			
Viewing surface brightness	3.8	4.13	0	0	3 each lot
Viewing surface image contrast	3.9	4.14	0	0	3 each lot
V/P operating test	3.10	4.15	0	0	100 percent

4.6.6 Inspection of preparation for delivery. The sampling and inspection of the preservation-packaging and interior package marking shall be in accordance with the groups A and B quality conformance inspection requirements of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of the applicable container specification and the marking requirements of MIL-STD-129.

4.7 Methods of examination and test.

4.7.1 Visual and mechanical examination. The V/P shall be examined to verify that materials, design and construction, and workmanship are in accordance with all requirements not involving test.

4.8 Film plane temperature (see 3.4.3). The V/P shall be tested in accordance with ANSI-PH3.46-1971, except ambient temperature shall be +115°F and film density shall be 1.3 or greater.

4.9 Humidity (see 3.4.4). The V/P shall be tested in accordance with method 507, procedure I of MIL-STD-810.

4.10 Resolution of fine detail (see 3.5). The V/P, when properly focused, with 24X lens or 48X lens, on a film target per figure 2, reduced 24X or 48X respectively, will resolve 3.2 lines/mm or higher on the viewing surface in all four corners and the center without refocusing. (Compromise focusing may be used.) The 24X and 48X reduction film target shall be certified to have a minimum resolving power of 108 lines/mm and 216 lines/mm respectively and a bar to space width tolerance of ±10 percent when viewed with a 100X or 200X microscope respectively.

4.11 Acoustic noise (see 3.6). The V/P shall meet the requirements of 3.6 when measured at 3 feet from the V/P on all four sides using a sound level meter complying with the requirements of ANSI-S1.4-1971.

4.12 Electromagnetic interference (EMI) (see 3.7). The V/P shall be tested to comply with MIL-STD-461, class IIB, and MIL-STD-462.

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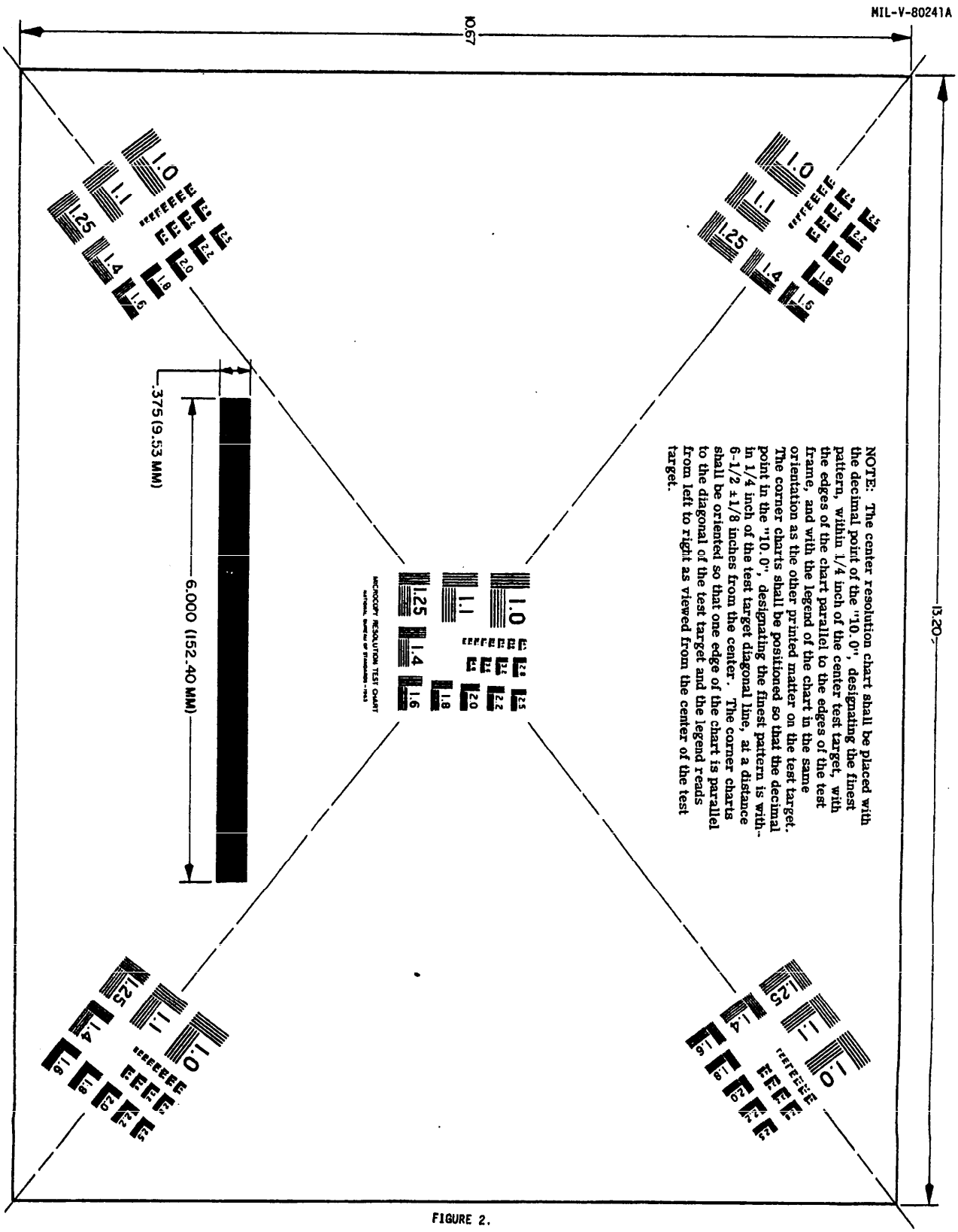


FIGURE 2.

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4.13 Viewing surface brightness (see 3.8). The V/P shall be tested with the methods prescribed in ANSI-PH5.10-1969. These methods shall apply to both front and rear projection viewers. All tests shall be performed with the platens (when used) in the optical path. The ANSI-PH5.10-1969 test will also be performed at a distance of 30 inches for first article test and at the option of the Government.

4.14 Viewing surface image contrast (see 3.9). The V/P shall be tested in accordance with the methods prescribed in ANSI-PH5.10-1969. These methods shall apply to both front and rear projection screens.

4.15 Viewer/printer operating test (see 3.10).

4.15.1 Viewer operating test. The V/P shall be put into operating condition by connecting the power plug into a power source. The test lamp shall be turned on. Each control shall be checked for correct operation. With the appropriate Government furnished fiche in the fiche carrier, the projection on the viewing surface shall be fully enlarged to the 24X and then the 48X magnification specified in 3.3.6.3. The viewing surface shall be examined for compliance with requirements herein. Any nonconformance in operation shall be cause for rejection of the viewer.

4.15.2 Printer operating test. Following the viewer operating test (see 4.15.1), a print will be produced; this print shall meet the requirements of 3.3.12.1.

4.16 Vibration (see 3.11). The V/P's shall be tested in accordance with MIL-STD-167, type I, except chemicals and lamp may be removed during test.

5. PREPARATION FOR DELIVERY

5.1 Preservation-packaging. Preservation-packaging shall be level A or C, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. V/P shall be cleaned in accordance with MIL-P-116, process C-1.

5.1.1.2 Drying. V/P shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative application. Preservatives shall not be used.

5.1.1.4 Unit packaging. V/P shall be individually packaged in accordance with MIL-P-116, submethod IA-14 insuring compliance with the general requirements paragraph under methods of preservation (unit protection) and the physical protection requirements paragraph therein. Hardware and accessories, such as maintenance manuals, accompanying each viewer shall be placed inside the unit package in a manner that will not damage the viewer. The unit container shall conform to PPP-B-636.

5.1.1.5 Intermediate packaging. Not required.

5.1.2 Level C. V/P shall be clean, dry and individually packaged in a manner that will afford adequate protection against corrosion, deterioration and physical damage during shipment from supply source to the first receiving activity.

5.2 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

5.2.1 Level A. The packaged V/P's shall be packed in fiberboard containers conforming to PPP-B-636, class weather resistant, style optional, special requirements. In lieu of the closure and waterproofing requirements in the appendix of PPP-B-636, closure and waterproofing shall be accomplished by sealing all seams, corners and manufacturer's joint with tape, two inches minimum width, conforming to PPP-T-60, class 1 of PPP-B-76. Banding (reinforcement requirements) shall be applied in accordance with the appendix to PPP-B-636 using nonmetallic or tape banding only. Unit packages with outer containers conforming to PPP-B-636, class weather resistant, special requirements require no further packing provided the requirements for closure, waterproofing and banding specified herein are met.

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5.2.2 Level B. The packaged V/P's shall be packed in fiberboard containers conforming to PPP-B-636, class domestic, style optional, special requirements. Closures shall be in accordance with the appendix thereto. Unit packages with outer containers conforming to PPP-B-636 require no further packing.

5.2.3 Level C. The packaged V/P's shall be packed in shipping containers in a manner that will afford adequate protection against damage during direct shipment from the supply source to the first receiving activity. These packs shall conform to the applicable carrier rules and regulations. Unit packages with outer containers conforming to PPP-B-636 require no further packing.

5.2.4 Unitized loads. Unitized loads, commensurate with the level of packing specified in the contract or order, shall be used whenever total quantities for shipment to one destination equal 40 cubic feet or more. Quantities less than 40 cubic feet need not be unitized. Unitized loads shall be uniform in size and quantities to the greatest extent practicable.

5.2.4.1 Level A. V/P's, packed as specified in 5.2.1, shall be unitized on pallets in conformance with MIL-STD-147, load type I, with a fiberboard cap (storage aid 4) positioned over the load.

5.2.4.2 Level B. V/P's, packed as specified in 5.2.2, shall be unitized as specified in 5.2.4.1 except that the fiberboard caps shall be class domestic.

5.2.4.3 Level C. V/P's, packed as specified in 5.2.3, shall be unitized with pallets and caps of the type, size and kind commonly used for this purpose and shall conform to the applicable carrier rules and regulations.

5.3 Marking. In addition to any marking required by the contract or purchase order (see 6.2), each unit package, exterior container and unitized load shall be marked in accordance with MIL-STD-129. Each exterior container shall be marked with fragile labels, arrows and the word "UP" as specified in MIL-STD-129.

5.4 Exterior containers. Exterior containers (see 5.2.1, 5.2.2, and 5.2.3) shall be of a minimum tare and cube consistent with the protection required and shall contain equal quantities of identical stock numbered items to the greatest extent practicable.

6. NOTES

6.1 Intended use. The V/P is to be used to enlarge microfiche images, to a readable size and produce a paper print of the enlarged image.

6.2 Ordering data.

(a) Levels of preservation-packaging and packing required (see 5.1 and 5.2).
(b) Special marking, if required (see 5.3).
(c) Cost per printout may be a bid evaluation factor for contract award (see 3.3.12).
(d) Magnification requirements; that is, 24X, 48X, or both (see 3.3.6.3).

6.3 First article inspection. Information pertaining to first article inspection of products covered by this specification should be obtained from the procuring activity for the specific contracts involved (see 3.1).

6.4 Definitions.

6.4.1 Fiche. A sheet of microfilm containing multiple microimages in a grid pattern.

6.4.2 Frame. A geometric subdivision of the fiche grid. That area of a COM produced fiche which contains microimage information, without margins.

6.4.3 Grid or grid pattern. A defined array of horizontal and vertical lines which devide the plane area of the fiche into uniform spaces called frames. The grid defines the arrangement of the frames into rows and columns of microimages.

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6.4.4 Image. Reproduction of printed matter created by an optical system.

6.4.5 Microform. A generic term for any form, either film or paper, which contains microimages.

6.4.6 Microimage. A unit of information, such as a page of text or a drawing, too small to be read without magnification.

6.4.7 Optical system. Lenses, mirrors, and lamp combination used in forming an image of a microform.

6.4.8 Viewer/printer. A projection device used for viewing a microimage and reproduce it on paper for reading with the unaided eye.

6.4.9 Resolution. The ability of optical systems and photo materials to make the fine detail of an object visible, expressed as number of lines discernible per millimeter.

6.5 Maintenance manual. Two copies of the manufacturer's service manual for maintenance and repair purposes should be provided for those addresses stipulated in the contract. This manual shall include a complete list of repair parts and a list of recommended running spares to support the initial year of operation.

6.6 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians:
Army - EL
Navy - SA
Air Force - 70

Review activities:
Army - MI, AT, ME, AV, WC
Navy - AS, EC, YD, OS, SH, MC
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User activities:
Army - SM, TM, IH, AD, WC, GL
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Preparing activity:
DSA - ES

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